

BARQ V1

Projections and Demonstration

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Abstract

BARQ V1 is a twelve-degree-of-freedom quadruped robot built on a proven 3D-printed chassis and driven by an NVIDIA Jetson Orin Nano. This brief assesses, candidly and strictly against the constraints of the existing hardware, what the platform can demonstrably do today and what it can be extended to do with no further hardware investment. The control software has been rebuilt from the ground up around a simulation-first methodology: every motion the robot performs is first validated in a physics-accurate digital twin and only then executed on hardware, using the identical motion commands. We report quantitative results from that twin — the robot stands, holds commanded postures to within a fraction of a degree, and walks a stable quasi-static gait — and we set out a concrete demonstration plan and a realistic forward roadmap. We are deliberate about the platform's defining characteristic: the actuators provide no position feedback, which bounds the robot to the quasi-static motion regime. The architecture is designed to exploit that regime fully and safely rather than to fail at its edges.

1 Introduction

BARQ V1 is the hardware-complete first generation of the BARQ quadruped program. The mechanical platform derives from the widely-replicated SpotMicro open-source design — a four-legged robot with three actuated joints per leg — adapted to carry an NVIDIA Jetson Orin Nano as its on-board computer. The present effort is a clean-sheet revival of the control software. The original stack, while functional in parts, was not maintainable; the platform has been rebuilt around rigorous engineering practice: a single audited source of measured hardware truth, a layered control stack, an automated test suite, and a physics simulation twin.

This document is a frank capabilities-and-constraints assessment intended for project review. It separates three things that are often conflated: what is **validated today**, what the **near-term demonstration** will show on the physical robot, and what is **realistically achievable later** on the very same hardware. Every claim of capability is tied either to a measured result or to a clearly-scoped engineering task.

1.1 System at a Glance

Subsystem	Specification
Configuration	12 DOF quadruped (4 legs $\times$ 3 joints)
Compute	NVIDIA Jetson Orin Nano (GPU-capable edge module)
Actuators	12 $\times$ DS3240MG — 40 kg $\cdot$ cm (3.92 N $\cdot$ m), 270° range
Actuation interface	PCA9685 PWM driver over I ² C, 50 Hz frame rate
Inertial sensing	MPU6050 IMU (accelerometer, gyroscope, magnetometer)
Vision	Luxonis OAK-D Pro — stereo depth, RGB, on-board neural inference
Power	4S LiPo battery (currently bench-tethered)
Mass / length	approx. 4.9 kg / 345 mm
Limb lengths	hip 55 mm, upper leg 107.5 mm, lower leg 130 mm

Table 1. BARQ V1 hardware configuration.

2 Engineering Methodology

The value of BARQ V1 to a reviewer lies as much in *how* it is being built as in what it does. Four principles govern the work and together de-risk every step from software to silicon.

2.1 A Single Source of Measured Truth

Every electrical and mechanical constant — servo pulse ranges, channel assignments, joint limits, link lengths — lives in one audited module and is never duplicated. If a number is wrong, there is exactly one place to correct it. This discipline is what failed in the first iteration and is now enforced.

2.2 Simulation-First Validation

A physics-accurate digital twin of the robot — its true geometry, mass distribution, and actuator torque and speed limits — runs in the PyBullet simulator. No motion is attempted on the physical robot until it has been shown stable in the twin. This converts the dangerous, hardware-damaging trial-and-error that ended the first iteration into cheap, repeatable software experiments.

2.3 Simulation–Hardware Parity

The simulator and the physical robot are driven by the *same* motion-command generator, at the same 50 Hz rate. A behaviour proven in simulation is not re-implemented for hardware — it is the identical command stream, sent to a real actuator instead of a simulated one. What is validated is literally what is executed.

2.4 A Permanent Safety Envelope

Mechanical-limit clamps, command slew-rate limiting, staggered actuator power-on, and an always-available emergency stop are built into the command path and remain there under every mode of operation. The robot cannot be commanded outside its safe mechanical range.

3 Validated Capabilities

The following results are measured in the simulation twin, which models the robot's real geometry and actuator limits. They constitute the regression baseline against which all hardware results will be compared.

Capability	Metric	Result
Inverse kinematics	round-trip geometric error	$< 10^{-6}$ m (15/15 tests)
Static posture tracking	RMS error, roll / pitch / yaw	0.31° / 0.15° / 0.11°
Standing stability	support-polygon margin	94 mm
Single-leg lift	tripod stability margin	33.5 mm (body tilt 0.7°)
Quasi-static walk (3 cycles)	distance / heading drift	91 mm / 0.59°
Walk stability	minimum support margin	12.8 mm (never unstable)
Real-time control	loop rate / timing overruns	50 Hz / 0

Table 2. Validated performance in the simulation twin.

3.1 Kinematics and Posing

An analytic inverse-kinematics engine maps any target body pose and foot placement to the twelve joint commands, verified to machine precision against its own forward model across the full reachable workspace. In practice this means the robot can be commanded to any feasible posture — lean, squat, raise or lower its body, look in a direction — with predictable, repeatable results.

3.2 Static Posture Control

With its feet planted, the robot tracks commanded body roll, pitch, yaw, and height to within roughly a third of a degree in simulation. This is the primitive underlying body stabilisation: the same control that holds a pose can be driven by the inertial sensor to keep the body level (see Section 6).

3.3 Quasi-Static Walking

BARQ V1 walks using a quasi-static crawl gait: at every instant at least three feet are on the ground and the centre of mass remains inside the resulting support triangle, so the robot is balanced geometrically rather than dynamically. Over three gait cycles in simulation the robot advanced 91 mm with only 0.59° of heading drift and never approached instability — the support margin stayed at or above 12.8 mm throughout. This is the platform's headline locomotion capability, and it is well within the envelope the hardware can support.

4 Demonstration Plan

The near-term demonstration reproduces the validated behaviours on the physical robot, in increasing order of capability. Each item maps directly to a measured result in Section 3, and the simulator provides a live side-by-side digital twin for any behaviour shown.

- **Calibration and posing.** The robot driven through its full posture envelope on a stand — leaning, squatting, and orienting its body — demonstrating precise, repeatable joint control.
- **Standing and body articulation.** Rising from rest to a stable stance, then articulating body roll, pitch, yaw, and height with the feet planted — including holding the body level while the support surface is tilted by hand.
- **Walking.** The quasi-static crawl gait carrying the robot forward across flat ground, straight and repeatably.
- **Teleoperation.** Live operator control of both posture and walking through a standard game controller.
- **Perception.** The on-board depth camera detecting and reacting to an obstacle or a person — for example halting, or orienting toward a detected subject.

5 Capabilities and Constraints

The defining characteristic of the hardware is **open-loop actuation**: the servos accept position commands but report nothing back — no joint angle, velocity, or torque. This single fact sets the platform's operating envelope, and we state it plainly rather than around it.

5.1 What the Hardware Enables

- Repeatable, calibrated posing across the full twelve-degree-of-freedom workspace.
- Static and quasi-static locomotion — gaits in which balance is maintained geometrically at every instant. The validated crawl gait is exactly this class of motion.
- Body-level closed-loop control using the inertial sensor, the one feedback channel that does exist, for posture and levelling.
- Vision-driven behaviour using the on-board camera, which is wholly independent of joint feedback and well matched to the Jetson's compute.

5.2 What the Hardware Bounds

- **No joint-level closed loop.** The robot cannot directly detect or correct a servo that fails to reach its commanded angle; such errors are observable only indirectly, at the body level, through the inertial sensor.
- **A quasi-static, not dynamic, motion regime.** Motions are executed slowly enough that momentum is negligible and balance is maintained by geometry. This is a deliberate and well-understood operating envelope.
- **No direct force sensing** at the joints, until rail-current monitoring is added (Section 6).

We regard this envelope as well matched to the platform's purpose: a robust, demonstrable, and safely-operable quadruped. The engineering is built to operate confidently inside it.

6 Future Scope on the Same Hardware

Each of the following is achievable on the existing hardware with software effort alone — no new components beyond a current sensor already procured. They are ordered roughly by effort.

6.1 Expanded Gait and Behaviour Library

Turning, sideways (lateral) crawling, and in-place rotation are direct variations of the validated quasi-static gait. A library of scripted behaviours — sit, lie down, stand, stretch — follows from the posing capability already demonstrated.

6.2 Body-Level Stabilisation

The inertial sensor can drive the posture controller to hold the body level and steady on a statically tilted surface, keeping the platform composed when it is set down off-level — a closed loop using the feedback the robot does have.

6.3 Contact and Load Awareness

A current sensor on the servo power rail (an INA260, already procured) provides an indirect measure of actuator load. This enables stall and collision detection and a basic reaction to felt resistance — a pragmatic, achievable substitute for the joint torque sensing the servos lack.

6.4 Vision-Guided Autonomy

The on-board depth-and-AI camera supports obstacle detection and avoidance on flat ground, person detection and following, and lightweight visual odometry for self-localisation — all of which run on the Jetson without external compute. Combined with the walking and levelling capabilities, these enable a supervised autonomous mode within a flat operating area.

7 Conclusion

BARQ V1 is a hardware-complete quadruped with a rebuilt, rigorously-validated control stack. Within the quasi-static envelope set by its open-loop actuators, it stands, articulates, and walks with quantified stability, and it is both controllable and extensible. The simulation-first methodology de-risks every step and makes the forward roadmap — a broader behaviour repertoire, body-level stabilisation, load awareness, and vision-guided autonomy — a matter of software on hardware that is already in hand. We are confident in both the demonstration we can give today and the trajectory we can deliver on the same platform.